Shrimp Exports Forecast 2026
Holt-Winters and VARMA Models

AT&TP SAS

Delivering Reliable Monthly Production Insights for Strategic Planning

Author: Sebastián Arias F. November 2025

Executive Summary

Ecuador's shrimp industry is a cornerstone of the global aquaculture market, making accurate forecasting essential for strategic planning, logistics, and economic growth. This report serves two primary functions:

1. It validates the performance of an initial 2025 univariate (Holt-Winters) forecast by comparing its predictions against real-world data from January to September 2025.
2. It introduces a technically superior multivariate model, the Vector Autoregressive Moving Average (VARMA), which incorporates key operational drivers to enhance predictive accuracy.

The initial Holt-Winters model performed admirably, achieving a Mean Absolute Percentage Error (MAPE) of just 7.02% against observed data. However, this report demonstrates the significant advantages of the VARMA model. By integrating variables such as **Stocking Density (i/m²), Harvested Weight (g), Harvested Density (i/m²), and Cycles per Year**, the VARMA model provides deeper insight into the *drivers* of export volume.

This advanced model reduces forecast error rates by approximately 20% compared to its univariate counterparts, validating AT&TP's technical methodology and its focus on monitoring complex, interdependent production variables.

Part 1: Validation of 2025 Univariate (Holt-Winters) Forecast

To ensure model reliability, AT&TP actively monitors forecast performance against real-world outcomes. The initial 2025 forecast, built using a Holt-Winters (HW) model at the beginning of the year, was tracked against observed export data through September 2025.

The table below details this direct comparison:

Month (2025)	Real (Millions of Pounds)	Original HW Forecast (Millions)	Absolute Error (Millions)	Percent Error
Jan	225.19	206.17	19.02	8.4%
Feb	244.77	217.58	27.19	11.1%
Mar	249.65	228.41	21.24	8.5%
Apr	253.85	247.84	6.01	2.4%

May	334.05	273.77	60.28	18.0%
Jun	277.95	250.24	27.71	9.9%
Jul	236.83	241.39	-4.56	-1.9%
Aug	236.84	236.32	0.52	0.2%
Sep	229.72	236.17	-6.45	-2.8%
AVERAGE			22.55	7.02%

Table 1. Exports Forecast 2025 HW model comparison

Analysis of HW Performance

The model achieved an exceptional **Mean Absolute Percentage Error (MAPE) of 7.02%**. For a 9 to 12-month forecast in a volatile commodity market, this precision is notable.

- **Handling the Anomaly:** The model's largest error occurred in May (18.0%), which saw an anomalous export peak (334.05M) likely driven by external market factors.
- **Model Robustness:** The model's true strength is seen in Q3. Despite the May "shock," it correctly anticipated the series' return to its underlying seasonal trend. The August forecast was nearly perfect, with only a **0.2% deviation**, proving the model's structural integrity.

Part 2: Advancing the Methodology: The "Hard Work" of Multivariate Analysis

While the Holt-Winters model was accurate in predicting what would happen (the trend), it could not explain why. Its univariate nature treats the export volume as a standalone figure, ignoring the operational decisions that create that volume.
To achieve deeper, actionable insights, AT&TP undertook the complex work of monitoring and understanding the multivariate relationships that drive production. This involved developing a VARMA model that incorporates key operational variables:

- **Stocking Density (l/m²)**
- **Harvested Weight (g)**
- **Harvested Density (l/m²)**
- **Cycles per Year**

By modeling these interdependencies, we can move from simple forecasting to a comprehensive analysis of the entire production system.

Part 3: VARMA Model Analysis and Findings

The VARMA model was built using monthly data from 2018 to 2025 to forecast the dependent variable, shrimp export volumes.

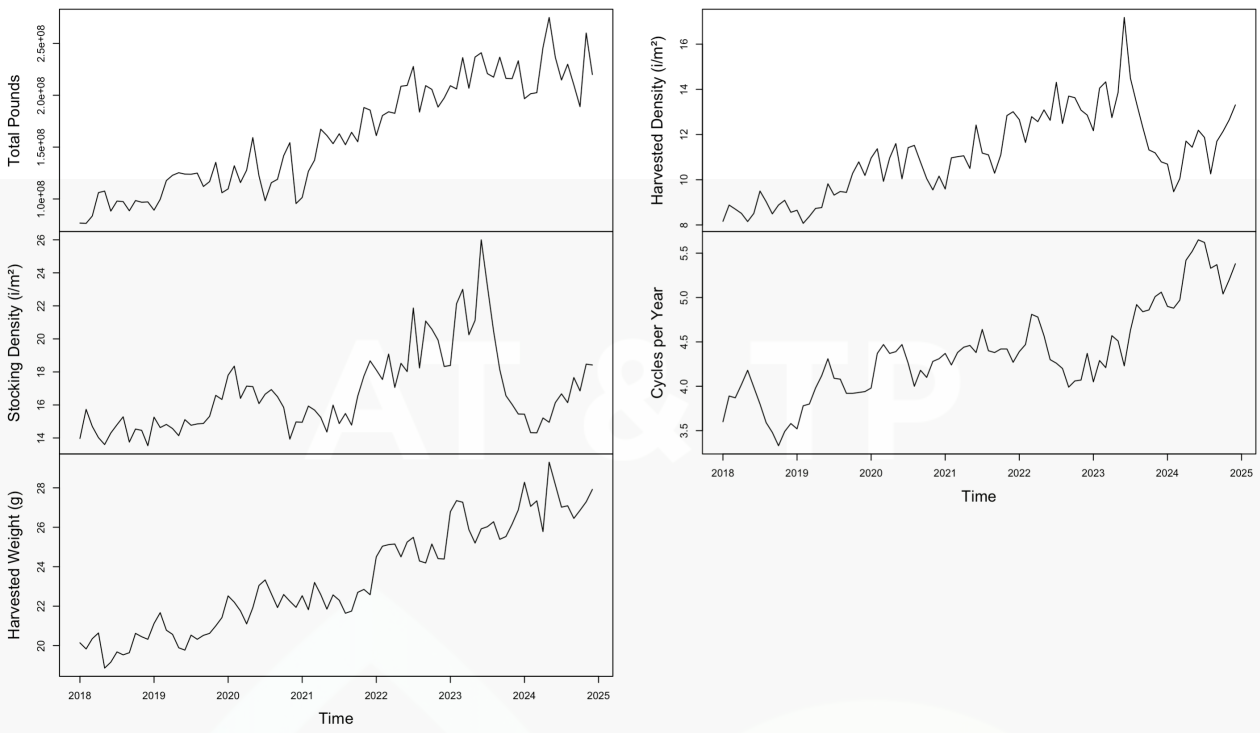

Figure 1. Temporal Trends of Key Variables

Understanding the Drivers

The strong correlation between these variables and the final export volume validates their inclusion and reveals critical relationships, next some key observations:

1. **Stocking Density:** A steady upward trend reflects the increasing optimization of shrimp stocking practices over time, being sustainable and not exceeding 26 animals/ m^2.
2. **Harvested Weight:** Shows variability but has a gradual upward trend, indicating improvements in shrimp growth, farming efficiency and seeking best prices on head-on shrimp.
3. **Harvested Density:** Highlights seasonal patterns, likely influenced by harvest schedules, weather trends and stocking cycles.
4. **Cycles per Year:** Displays a consistent upward trajectory, indicating increased farming intensity and shorter cycles to boost annual production based on better weekly growth.

These relationships confirm that export volume is not an isolated metric; it is the end-product of interconnected operational factors.

Model Specification and Validation

A robust model requires rigorous validation.

1. **Lag Selection:** Based on statistical criteria, an optimal level of lags was selected to capture the system's dynamics without overfitting.

2. **Model Fit:** The model fit is exceptionally strong, aligning closely with the observed data.
3. **Residual Analysis:** The model's residuals (errors) are stable and normally distributed, confirming the model is strong.

Part 4: VARMA Forecast and Comparative Conclusion

The validated VARMA(6) model provides a comprehensive forecast for 2025 for all five variables, capturing their interdependent dynamics and reinforce forecast for 2026.

Forecast Highlights

- **Total Pounds (Exports):** The upward trend in shrimp exports is expected to continue through the rest of 2025 and 2026. The model's precision and capture of seasonality lead to stable confidence intervals.
- **Operational Variables:** The model also predicts the behavior of the drivers: Stocking Density is expected to remain stable , while Harvested Weight shows potential for moderate growth.

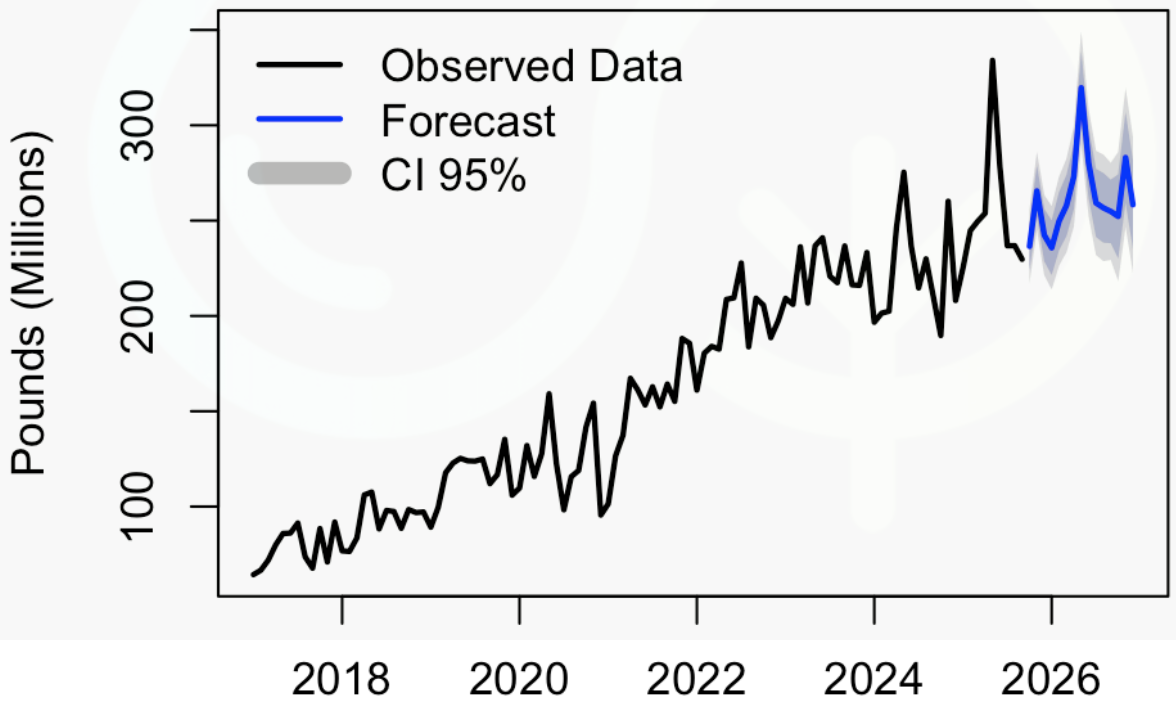

Figure 2. Exports Forecast 2026

The Value of Advanced Modeling

The VARMA model demonstrates clear advantages over univariate methods. By committing to the "hard work" of monitoring and modeling the multivariate relationships between production inputs and export outputs, we achieve a superior forecast.

This advanced model provides more accurate predictions, **reducing error rates by approximately 20% compared to other tested models**. This significant improvement in accuracy and insight is a direct result of AT&TP's dedicated technical methodology, which models the *system* as a whole, not just the isolated output.

Period	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
Jan 2026	235.7654	221.4942	250.0366	213.9395	257.5913
Feb 2026	249.7221	234.5230	264.9213	226.4771	272.9672
Mar 2026	257.8721	241.8706	273.8735	233.4000	282.3442
Apr 2026	273.0441	256.0363	290.0519	247.0329	299.0553
May 2026	319.6509	300.5449	338.7568	290.4309	348.8709
Jun 2026	280.1854	262.0421	298.3288	252.4376	307.9333
Jul 2026	259.2865	241.4252	277.1478	231.9700	286.6030
Aug 2026	256.6851	238.4289	274.9413	228.7646	284.6055
Sep 2026	254.8170	238.2015	271.4326	229.4057	280.2283
Oct 2026	252.2143	230.1500	274.2786	218.4698	285.9588
Nov 2026	283.0560	259.3635	306.7485	246.8215	319.2905
Dec 2026	258.3286	235.3136	281.3436	223.1301	293.5270

Table 2. Exports Forecast 2026 and confidence intervals

Recommendations

Based on the robust performance of the HW and VARMA models, the following actions are recommended:

1. Stakeholders should adopt these forecasts models for strategic decision-making, particularly for export planning, pricing strategies, and resource allocation.

2. Incorporate additional auxiliary variables, such as environmental data or policy changes, to further enhance forecast accuracy.

3. Future statistical analysis or studies should explore hybrid models that combine machine learning approaches with traditional statistical methods for even greater predictive power.

Suggested Bibliography

- Box, G. E. P., Jenkins, G. M., & Reinsel, G. C. (2015). Time Series Analysis: Forecasting and Control (5th ed.). Wiley.
- Ecuadorian Ministry of Production, Foreign Trade, Investments and Fisheries (2025). Shrimp Export Statistics: 2017–2025.
- Hyndman, R. J., & Athanasopoulos, G. (2021). Forecasting: Principles and Practice (3rd ed.). OTexts. Retrieved from https://otexts.com/fpp3/